Q1.
A file of size **12,300 bytes** is stored on a disk with **block size = 512 bytes**.
How many blocks are required?
If the FAT entry size is **4 bytes per block**, how much space will FAT use?

Solution

Number of blocks = ⌈ file size / block size ⌉
= ⌈ 12,300 / 512 ⌉
= ⌈ 24.02 ⌉
= **25 blocks**

FAT size = number of blocks × size of each entry
= 25 × 4 = **100 bytes**

✓ Blocks required = 25
✓ FAT space used = 100 bytes

Q2.
Block size = **1 KB**
Each index entry = **4 bytes**
How many data blocks can a single-level index block point to?

Solution

Index block capacity = block size / entry size
= 1024 / 4
= **256 entries**

✓ Index block points to 256 data blocks

Max file size = 256 × 1 KB = **256 KB**

Q3.
Block size = **1024 bytes**
Index entry = **4 bytes**

A file uses:

- 12 direct pointers
- 1 single indirect pointer
- 1 double indirect pointer

Find **maximum file size**.

Solution

Direct blocks

= 12 blocks = 12 × 1024 = **12 KB**

Single indirect

Entries in 1 block = 1024/4 = 256 blocks
→ 256 × 1 KB = **256 KB**

Double indirect

One indirect block = 256 entries
Each points to another indirect block → 256 × 256 data blocks

= 256^2 = 65,536 blocks
→ 65,536 × 1 KB = **65,536 KB** = **64 MB**

Total max file size

= 12 KB + 256 KB + 64 MB
≈ **64.25 MB**

✓ Maximum file size ≈ 64.25 MB

Q4.
Disk block size = **4 KB**
File size = **18 KB**
Stored as contiguous blocks.
Find internal fragmentation.

Solution

Blocks required = ⌈18 / 4⌉ = ⌈4.5⌉ = **5 blocks**
Total allocated = 5 × 4 = **20 KB**

Internal fragmentation = allocated – used
= 20 – 18 = **2 KB**

✓ Internal fragmentation = 2 KB